Sensitivity analysis

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Outline

Optimal bases and their two conditions

Changes in the objective-function vector
- Objective-function coefficient of a non-basic variable
- Objective-function coefficient of a basic variable

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Adding a new variable

Adding a new constraint

Optimal bases and their two conditions

Introduction: Optimal bases and their two conditions

- In real applications the data of an LP problem are **estimates**: costs, prices and resources may change after the problem has been solved
- Solving the LP problem again from scratch at every change is wasteful: the **sensitivity analysis** reuses the optimal basis already found
- The starting point is the definition of an **optimal basis** through two conditions, primal feasibility and dual feasibility
- We also introduce the running example of the chapter, an LP problem with $n = 2$ variables and $m = 3$ constraints

Sensitivity analysis

- The input data of an LP problem are rarely known exactly
- The entries of the objective-function vector $\boldsymbol{c}$ and of the right-hand-side vector $\boldsymbol{b}$ are **estimates**, and they may change after the LP problem has been solved
- The **sensitivity analysis** studies how an optimal solution and the optimal value change when the input data change

The LP problem in standard maximization form

- We consider an LP problem in standard maximization form with $n + m$ non-negative variables and m equality linear constraints
- It is obtained from an LP problem in canonical maximization form by adding m slack variables:

$$\begin{aligned} \max \quad & \boldsymbol{c}' \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0. \end{aligned}$$

Optimal basis

Definition 1: optimal basis

*A basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard maximization form is called an **optimal basis** if it satisfies the following two conditions, and then its basic feasible solution is an optimal solution:*

1. ***primal feasibility**:*

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \geq 0;$$

2. ***dual feasibility**:*

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\, \boldsymbol{A}_{\mathcal{N}} \leq 0, \qquad \textit{where} \quad \boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}.$$

Two sufficient conditions

- The two conditions are sufficient for optimality
- Under degeneracy, an optimal basic feasible solution may also be associated with a basis which is not dual feasible
- A change of the input data preserves the optimality of the basis as long as both conditions keep holding

Which data appear in which condition

The two conditions involve the input data in two different ways:

1. the right-hand-side vector $\boldsymbol{b}$ appears only in the primal feasibility condition;
2. the objective-function vector $\boldsymbol{c}$ appears only in the dual feasibility condition.

- Accordingly, a change of $\boldsymbol{b}$ may only destroy the **primal** feasibility of the basis
- A change of $\boldsymbol{c}$ may only destroy its **dual** feasibility

Let us now fix the running example of the chapter: an LP problem with its optimal basis, its dictionary and the matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$.

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Example: optimal basis (1/4)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrcrcrr}
\max & 5 & x_1 & + & 2 & x_2 & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

Example: optimal basis (2/4)

- Its standard maximization form has $n + m = 5$ non-negative variables
- The optimal basis determined by the simplex algorithm is

$$\mathcal{B} = \{1, \ 2, \ 3\} \qquad \text{and} \qquad \mathcal{N} = \{4, \ 5\}.$$

Example: optimal basis (3/4)

The associated dictionary is:

$$\begin{cases} x_1 = 5 - \frac{1}{4}x_4 - \frac{1}{4}x_5 \\ x_2 = \frac{7}{2} - \frac{3}{8}x_4 + \frac{1}{8}x_5 \\ x_3 = \frac{13}{2} + \frac{7}{8}x_4 - \frac{5}{8}x_5 \\ z = 32 - 2x_4 - x_5 \end{cases}$$

- All the reduced costs are non-positive and all the values of the basic variables are non-negative: the basis is optimal

Example: optimal basis (4/4)

We have:

$$\boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{3}{8} & -\frac{1}{8} \\ 1 & -\frac{7}{8} & \frac{5}{8} \end{pmatrix} \qquad \text{and} \qquad \boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1} = (0 \;\; 2 \;\; 1).$$

An optimal solution is

$$\tilde{\boldsymbol{x}}(\, LP\,) = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$$

and the optimal value is $z(\, LP\,) = 32$.

Summary

Optimal basis:

- Primal feasibility: $\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b} \geq 0$
- Dual feasibility: $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\,\boldsymbol{A}_{\mathcal{N}} \leq 0$, with $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\,\boldsymbol{A}_{\mathcal{B}}^{-1}$
- The two conditions are sufficient for optimality

Data and conditions:

- A change of $\boldsymbol{b}$ may only destroy primal feasibility
- A change of $\boldsymbol{c}$ may only destroy dual feasibility

Running example: $\mathcal{B} = \{1,\ \ 2,\ \ 3\}$, $\boldsymbol{\pi} = (0\ \ 2\ \ 1)$, $z(\,LP\,) = 32$

Changes in the objective-function vector

Introduction: Changes in the objective-function vector

- A cost or a profit changes: for which values does the optimal basis remain optimal?
- Only the **dual feasibility** condition is involved: we compute an interval of values for the changed coefficient
- Two cases: the coefficient of a **non-basic** variable, which changes one reduced cost, and the coefficient of a **basic** variable, which changes all of them
- On the running example we compute the intervals and we see their geometric meaning: the level sets of the objective function rotate

Changing one entry of $\boldsymbol{c}$

- We replace one entry c_k of the objective-function vector with

$$c_k + \delta, \qquad \text{where } \delta \in \mathbb{R}$$

- We look for the values of δ for which the optimal basis remains optimal
- Since $\boldsymbol{b}$ does not change, the primal feasibility condition keeps holding
- Only the **dual feasibility** condition has to be checked

Coefficient of a non-basic variable

- If $k \in \mathcal{N}$, then $\boldsymbol{c}_{\mathcal{B}}$ does not change, and then $\boldsymbol{\pi}$ does not change either
- Only the reduced cost of x_k changes:

$$\bar{c}_k \longrightarrow \bar{c}_k + \delta$$

- The optimal basis remains optimal if and only if

$$\delta \leq -\bar{c}_k$$

Non-basic variable: solution and value

- The optimal solution and the optimal value do not change for these values of δ
- The dictionary is the same, except for one reduced cost

For a basic variable the situation is different:
its coefficient appears in $\boldsymbol{c}_{\mathcal{B}}$, and then in $\boldsymbol{\pi}$.

Coefficient of a basic variable (1/2)

- If $k \in \mathcal{B}$, then $\boldsymbol{c}_{\mathcal{B}}$ changes, and then all the reduced costs change
- We denote by $\bar{a}_{kj}$ the entry of the matrix $\bar{\boldsymbol{A}}$ in the row of x_k and in the column of x_j
- We have

$$\bar{c}_j \longrightarrow \bar{c}_j - \delta\, \bar{a}_{kj}, \qquad \forall j \in \mathcal{N}$$

Coefficient of a basic variable (2/2)

- The optimal basis remains optimal if and only if

$$\bar{c}_j - \delta\, \bar{a}_{kj} \leq 0, \qquad \forall j \in \mathcal{N}$$

- That is, if and only if

$$\delta \geq \max \left\{ \frac{\bar{c}_j}{\bar{a}_{kj}} : j \in \mathcal{N},\ \bar{a}_{kj} > 0 \right\}$$

$$\delta \leq \min \left\{ \frac{\bar{c}_j}{\bar{a}_{kj}} : j \in \mathcal{N},\ \bar{a}_{kj} < 0 \right\}$$

- A positive entry $\bar{a}_{kj}$ gives a **lower bound** on δ, a negative one an **upper bound**

Basic variable: solution and value

- The optimal solution does not change for these values of δ, since $\bar{\boldsymbol{b}}$ is unchanged
- The optimal value, instead, becomes

$$z(\,LP\,) + \delta\ \tilde{x}_k$$

Let us compute the intervals of the two basic variables x_1 and x_2 of the running example.

Example: changes in the objective-function vector (1/6)

Consider the optimal basis of the running example, with

$$\mathcal{N} = \{4, \ \ 5\}, \qquad \bar{c}_4 = -2, \qquad \bar{c}_5 = -1$$

and

$$\bar{\boldsymbol{A}} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{3}{8} & -\frac{1}{8} \\ -\frac{7}{8} & \frac{5}{8} \end{pmatrix}.$$

- The rows of $\bar{\boldsymbol{A}}$ correspond to the basic variables x_1, x_2, x_3, and its columns to the non-basic variables x_4, x_5

Example: changes in the objective-function vector (2/6)

Coefficient of the basic variable x_1. We replace $c_1 = 5$ with $5 + \delta$.

- The row of x_1 in $\bar{\boldsymbol{A}}$ is $\begin{pmatrix} \frac{1}{4} & \frac{1}{4} \end{pmatrix}$, whose two entries are both positive, and then both of them give a lower bound on δ:

$$\delta \geq \max \left\{ \underbrace{\frac{-2}{1/4}}_{=-8}, \underbrace{\frac{-1}{1/4}}_{=-4} \right\} = -4$$

- There is no upper bound, since no entry of the row is negative. Accordingly,

$$\delta \geq -4 \qquad \Longrightarrow \qquad c_1 \in [\,1,\, +\infty\,).$$

Example: changes in the objective-function vector (3/6)

Coefficient of the basic variable x_2. We replace $c_2 = 2$ with $2 + \delta$.

- The row of x_2 in $\bar{\boldsymbol{A}}$ is $\begin{pmatrix} \frac{3}{8} & -\frac{1}{8} \end{pmatrix}$, with one positive entry, which gives a lower bound, and one negative entry, which gives an upper bound:

$$\underbrace{\frac{-2}{3/8}}_{=-\frac{16}{3}} \leq \delta \leq \underbrace{\frac{-1}{-1/8}}_{=8}$$

- That is,

$$-\frac{16}{3} \leq \delta \leq 8 \qquad \Longrightarrow \qquad c_2 \in \left[-\frac{10}{3},\ 10 \right].$$

Example: changes in the objective-function vector (4/6)

- For every value of c_2 in this interval, the optimal solution remains

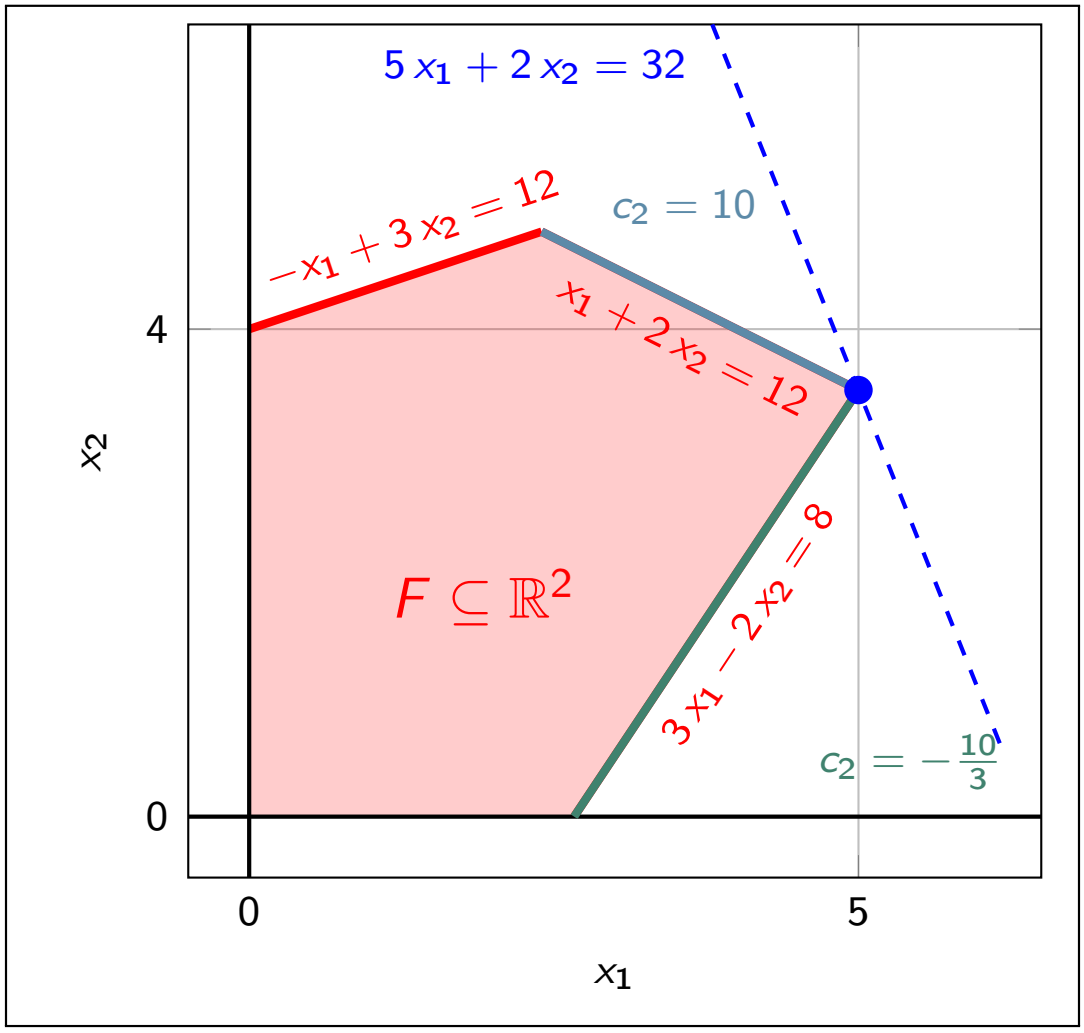
$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$$

- The optimal value becomes

$$32 + \delta \cdot \frac{7}{2}$$

- In the next slide: the feasible region F in red, the level set $5\,x_1 + 2\,x_2 = 32$ (dashed), and the two edges meeting at the optimal vertex, in green and in blue

Example: changes in the objective-function vector (5/6)



Example: changes in the objective-function vector (6/6)

- The level sets of the objective function **rotate** as c_2 varies, while the optimal solution remains the vertex $\begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$
- The two endpoints of the interval are the values of c_2 for which the level sets become parallel to one of the two edges meeting at that vertex:
  - for $c_2 = 10$ they are parallel to the edge on $x_1 + 2x_2 = 12$, drawn in blue;
  - for $c_2 = -\frac{10}{3}$ they are parallel to the edge on $3x_1 - 2x_2 = 8$, drawn in green
- Beyond those values the optimal solution moves to an adjacent vertex

Summary

Change $c_k \to c_k + \delta$: only dual feasibility has to be checked

Non-basic variable ($k \in \mathcal{N}$):

- Only $\bar{c}_k$ changes: the basis remains optimal iff $\delta \leq -\bar{c}_k$
- Optimal solution and optimal value do not change

Basic variable ($k \in \mathcal{B}$):

- All reduced costs change: $\bar{c}_j \to \bar{c}_j - \delta\, \bar{a}_{kj}$
- Interval of δ from the ratios $\bar{c}_j / \bar{a}_{kj}$ of the row of x_k
- The optimal solution does not change, the optimal value becomes $z(\,LP\,) + \delta\, \tilde{x}_k$
- Geometrically, the level sets rotate around the optimal vertex

Changes in the right-hand-side vector

Introduction: Changes in the right-hand-side vector

- An available resource changes: for which values does the optimal basis remain optimal?
- Only the **primal feasibility** condition is involved: the values of the basic variables move linearly with the change
- We introduce the **shadow prices**: they tell how much the optimal value gains for one more unit of a resource
- On the running example we compute the intervals of the three right-hand sides and we see the constraint translate

Changing one entry of $\boldsymbol{b}$

- We replace one entry b_k of the right-hand-side vector with

$$b_k + \delta, \qquad \text{where } \delta \in \mathbb{R}$$

- Since $\boldsymbol{c}$ does not change, the dual feasibility condition keeps holding
- Only the **primal feasibility** condition has to be checked

Values of the basic variables

- The values of the basic variables become

$$\bar{\boldsymbol{b}} \longrightarrow \boldsymbol{A}_{\mathcal{B}}^{-1}\left(\boldsymbol{b}+\delta\,\boldsymbol{e}_k\right)=\bar{\boldsymbol{b}}+\delta\,\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{e}_k$$

- $\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{e}_k$ is the k-th column of the matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$
- We denote its entries by β_{ik}

Range of δ

- The optimal basis remains optimal if and only if

$$\bar{b}_i + \delta\, \beta_{ik} \geq 0, \qquad \forall i \in \mathcal{B}$$

- That is, if and only if

$$\delta \geq \max \left\{ -\frac{\bar{b}_i}{\beta_{ik}} : i \in \mathcal{B},\ \beta_{ik} > 0 \right\}$$

$$\delta \leq \min \left\{ -\frac{\bar{b}_i}{\beta_{ik}} : i \in \mathcal{B},\ \beta_{ik} < 0 \right\}$$

What happens to the optimal value in this interval?
The answer is given by the vector π.

Shadow prices

Definition 2: shadow prices

Given an optimal basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard maximization form, the entries of the row vector

$$\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1}$$

*are called the **shadow prices** of the m linear constraints.*

Shadow prices as marginal values

- As long as the optimal basis remains optimal, the optimal value changes linearly with δ:

$$z(\,LP\,) \;\longrightarrow\; z(\,LP\,) + \delta\,\pi_k$$

- The shadow price π_k is then the **marginal value** of the k-th constraint

- It measures the increase of the optimal value per unit increase of b_k

Shadow prices and duality

- By the Strong Duality theorem, the vector $\boldsymbol{\pi}$ of the shadow prices is an optimal solution of the dual problem, and then

$$z(\,LP\,) = \boldsymbol{\pi}\ \boldsymbol{b} = \sum_{i=1}^{m} \pi_i\ b_i$$

- If the k-th constraint is not binding at the optimal solution, i.e., if its slack variable is basic with a strictly positive value, then

$$\pi_k = 0$$

and a small change of b_k does not change the optimal value

Let us compute the intervals of the three right-hand sides b_1, b_2 and b_3 of the running example.

Example: changes in the right-hand-side vector (1/6)

Consider the optimal basis of the running example, with

$$\bar{\boldsymbol{b}} = \begin{pmatrix} 5 & \frac{7}{2} & \frac{13}{2} \end{pmatrix}' \qquad \text{and} \qquad \boldsymbol{\pi} = (0 \;\; 2 \;\; 1).$$

First constraint.

- Its shadow price is $\pi_1 = 0$, since its slack variable x_3 is basic with value $\frac{13}{2} > 0$
- The first constraint is not binding at the optimal solution, and a small change of b_1 does not change the optimal value
- The first column of $\boldsymbol{A}_{\mathcal{B}}^{-1}$ is $(0 \;\; 0 \;\; 1)'$, and then only the value of x_3 changes, giving

$$\frac{13}{2} + \delta \geq 0 \quad \Longrightarrow \quad \delta \geq -\frac{13}{2} \quad \Longrightarrow \quad b_1 \in \left[\frac{11}{2}, \, +\infty \right).$$

Example: changes in the right-hand-side vector (2/6)

Second constraint. Its shadow price is $\pi_2 = 2$.

The second column of $\boldsymbol{A}_{\mathcal{B}}^{-1}$ is $\left(\frac{1}{4} \quad \frac{3}{8} \quad -\frac{7}{8}\right)'$, and then

$$\begin{cases} 5 + \frac{1}{4}\,\delta \geq 0 \\ \frac{7}{2} + \frac{3}{8}\,\delta \geq 0 \\ \frac{13}{2} - \frac{7}{8}\,\delta \geq 0 \end{cases} \implies \begin{cases} \delta \geq -20 \\ \delta \geq -\frac{28}{3} \\ \delta \leq \frac{52}{7} \end{cases} \implies -\frac{28}{3} \leq \delta \leq \frac{52}{7}$$

that is,

$$b_2 \in \left[\frac{8}{3},\ \frac{136}{7}\right].$$

Example: changes in the right-hand-side vector (3/6)

- For every value of b_2 in this interval, the optimal value is

$$z(\,LP\,) = 32 + 2\,\delta$$

- For instance, with $b_2 = 13$ we have $\delta = 1$ and the optimal value becomes 34

- With $b_2 = 11$ we have $\delta = -1$ and the optimal value becomes 30

Third constraint.

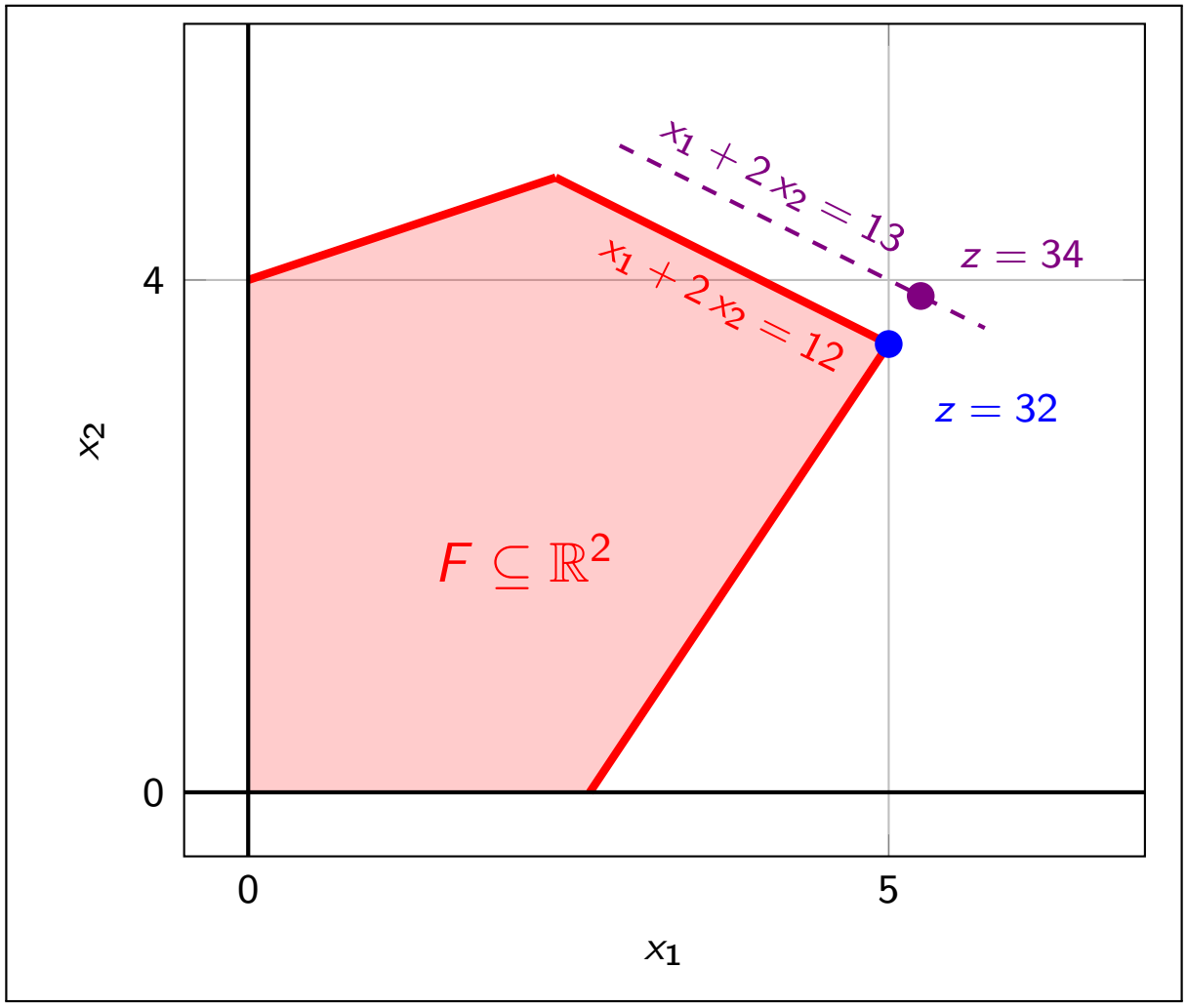
- Its shadow price is $\pi_3 = 1$, and its column is $\begin{pmatrix} \frac{1}{4} & -\frac{1}{8} & \frac{5}{8} \end{pmatrix}'$, which gives

$$b_3 \in \left[-\frac{12}{5},\ 36 \right].$$

Example: changes in the right-hand-side vector (4/6)

- We now look at the change $b_2 = 12 \rightarrow 13$ in the plane (x_1, x_2)
- In the next slide: the feasible region F in red, the constraint $x_1 + 2\,x_2 = 12$ and the translated constraint $x_1 + 2\,x_2 = 13$ (dashed, in violet)
- The two points are the optimal solutions before (blue, $z = 32$) and after (violet, $z = 34$) the change

Example: changes in the right-hand-side vector (5/6)



Example: changes in the right-hand-side vector (6/6)

- By increasing b_2 from 12 to 13, the second constraint **translates** outwards
- The optimal solution moves from

$$\begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} \frac{21}{4} \\ \frac{31}{8} \end{pmatrix}$$

- The optimal value increases by $\pi_2 = 2$

Summary

Change $b_k \to b_k + \delta$: only primal feasibility has to be checked

- $\bar{\boldsymbol{b}} \to \bar{\boldsymbol{b}} + \delta\, \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{e}_k$: the k-th column of $\boldsymbol{A}_{\mathcal{B}}^{-1}$, with entries β_{ik}
- Interval of δ from the ratios $-\bar{b}_i / \beta_{ik}$

Shadow prices $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}$:

- In the interval, $z(\,LP\,) \to z(\,LP\,) + \delta\, \pi_k$: π_k is the marginal value of the k-th constraint
- $\boldsymbol{\pi}$ is an optimal dual solution and $z(\,LP\,) = \boldsymbol{\pi}\, \boldsymbol{b}$
- A non-binding constraint has shadow price 0
- Geometrically, the constraint translates and the optimal vertex moves

Adding a new variable

Introduction: Adding a new variable

- A new activity becomes available, for instance a new product: is it convenient to use it?
- The old optimal basis is still **primal feasible**: only the reduced cost of the new variable has to be computed
- If the new variable is convenient, the **simplex algorithm** restarts from the old basis instead of from scratch
- On the running example a single pivoting operation is enough

A new variable

- We add one non-negative variable $x_\star$ to the LP problem in standard maximization form
- Its objective-function coefficient is $c_\star$ and its column is $\boldsymbol{A}_\star \in \mathbb{R}^m$
- The optimal basis of the original LP problem is still a basis of the new one
- It is still primal feasible, since neither $\boldsymbol{A}_\mathcal{B}$ nor $\boldsymbol{b}$ has changed

Reduced cost of the new variable

- The new variable $x_\star$ is non-basic, and its reduced cost is

$$\bar{c}_\star = c_\star - \boldsymbol{\pi}\, \boldsymbol{A}_\star$$

1. If $\bar{c}_\star \leq 0$, then the basis is still dual feasible and the optimal solution does not change: the new variable takes the value $\tilde{x}_\star = 0$

2. If $\bar{c}_\star > 0$, then the basis is no longer dual feasible: the **simplex algorithm** is restarted from it, with $x_\star$ as the first pivot column

Let us add a new variable to the running example
and restart the simplex algorithm from its optimal basis.

Example: adding a new variable (1/5)

Consider the LP problem of the running example and add a new non-negative variable $x_\star$ with objective-function coefficient $c_\star = 4$ and column $\boldsymbol{A}_\star = (1 \;\; 1 \;\; 1)'$, that is, the LP problem becomes:

$$
\begin{array}{rrrrrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & + & 4 & x_\star & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & + & & x_\star & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & + & & x_\star & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & + & & x_\star & \leq & 8 \\
 & & x_1, & & & x_2, & & & x_\star & \geq & 0
\end{array}
$$

Example: adding a new variable (2/5)

- The reduced cost of the new variable is

$$\bar{c}_\star = c_\star - \boldsymbol{\pi}\, \boldsymbol{A}_\star = 4 - (0 \;\; 2 \;\; 1) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 4 - 3 = 1 > 0$$

- The basis $\mathcal{B} = \{1, \;\; 2, \;\; 3\}$ is no longer dual feasible
- It is convenient to produce the new variable

Example: adding a new variable (3/5)

- We restart the simplex algorithm from this basis, with $x_\star$ as pivot column
- Its column in the dictionary is

$$\bar{\boldsymbol{A}}_\star = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_\star = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{3}{8} & -\frac{1}{8} \\ 1 & -\frac{7}{8} & \frac{5}{8} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{4} \\ \frac{3}{4} \end{pmatrix}$$

Example: adding a new variable (4/5)

- The ratio test gives

$$t_m = \min\left\{ \frac{5}{1/2}, \frac{7/2}{1/4}, \frac{13/2}{3/4} \right\} = \min\left\{ 10, 14, \frac{26}{3} \right\} = \frac{26}{3}$$

- The minimum is attained by the basic variable x_3, which **leaves** the basis

- The new variable $x_\star$ **enters** the basis

Example: adding a new variable (5/5)

- After this single pivoting operation all the reduced costs become non-positive
- The new optimal solution is

$$\tilde{x}_1 = \frac{2}{3}, \qquad \tilde{x}_2 = \frac{4}{3}, \qquad \tilde{x}_\star = \frac{26}{3}$$

- The optimal value is

$$z(\,LP\,) = 5 \cdot \frac{2}{3} + 2 \cdot \frac{4}{3} + 4 \cdot \frac{26}{3} = \frac{122}{3}$$

Summary

New variable $x_\star$ with coefficient $c_\star$ and column $\boldsymbol{A}_\star$:

- The old optimal basis is still primal feasible
- Reduced cost $\bar{c}_\star = c_\star - \boldsymbol{\pi}\,\boldsymbol{A}_\star$

Two cases:

- $\bar{c}_\star \leq 0$: the optimal solution does not change, $\tilde{x}_\star = 0$
- $\bar{c}_\star > 0$: the simplex algorithm restarts from the old basis, with $x_\star$ as the first pivot column

Example: $\bar{c}_\star = 1$, one pivoting operation, $z(\,LP\,) = \frac{122}{3}$

Adding a new constraint

Introduction: Adding a new constraint

- A new requirement appears, for instance a new limited resource: does the old optimal solution still satisfy it?
- The old optimal basis, extended with the new slack variable, is still **dual feasible**
- If the old optimal solution violates the new constraint, the **dual simplex algorithm** restarts from the extended basis
- On the running example a single dual pivoting operation is enough

A new constraint

- We add one linear constraint

$$\sum_{j=1}^{n+m} a_{\star j}\, x_j \leq b_\star$$

  to the LP problem in standard maximization form, together with its slack variable $x_{n+m+1} \geq 0$

- The optimal basis of the original LP problem, extended with x_{n+m+1}, is a basis of the new LP problem

The extended basis

- The extended basis is still dual feasible, since the reduced costs do not change and the reduced cost of the new slack variable is 0
- Its primal feasibility, instead, depends on the optimal solution $\tilde{\boldsymbol{x}}(\, LP\,)$ of the original LP problem:

1. if $\tilde{\boldsymbol{x}}(\, LP\,)$ **satisfies** the new constraint, then the extended basis is still optimal and the optimal solution does not change;
2. if $\tilde{\boldsymbol{x}}(\, LP\,)$ **violates** the new constraint, then the new slack variable takes a negative value and the **dual simplex algorithm** is restarted from the extended basis, with x_{n+m+1} as the first pivot row.

Let us add a new constraint to the running example
and restart the dual simplex algorithm from the extended basis.

Example: adding a new constraint (1/6)

- Consider the LP problem of the running example and add the new linear constraint

$$x_1 + x_2 \leq 6,$$

with its slack variable x_6

- The optimal solution of the original LP problem violates it, since

$$\tilde{x}_1 + \tilde{x}_2 = 5 + \frac{7}{2} = \frac{17}{2} > 6$$

- The value of the new slack variable is

$$\bar{b}_6 = 6 - \frac{17}{2} = -\frac{5}{2} < 0$$

Example: adding a new constraint (2/6)

- The extended basis $\mathcal{B} = \{1, \ 2, \ 3, \ 6\}$ is dual feasible, since the reduced costs are still $\bar{c}_4 = -2$ and $\bar{c}_5 = -1$
- It is not primal feasible, since $\bar{b}_6 < 0$
- We apply the **dual simplex algorithm**, with x_6 as pivot row
- The row of x_6 in the dictionary is obtained by substituting the equations of x_1 and x_2 in the new constraint

Example: adding a new constraint (3/6)

The row of x_6:

$$\begin{aligned} x_6 &= 6 - x_1 - x_2 \\ &= 6 - \left(5 - \frac{1}{4}\,x_4 - \frac{1}{4}\,x_5\right) - \left(\frac{7}{2} - \frac{3}{8}\,x_4 + \frac{1}{8}\,x_5\right) \\ &= -\frac{5}{2} + \frac{5}{8}\,x_4 + \frac{1}{8}\,x_5, \end{aligned}$$

that is,

$$\bar{a}_{64} = -\frac{5}{8} \qquad \text{and} \qquad \bar{a}_{65} = -\frac{1}{8}.$$

Example: adding a new constraint (4/6)

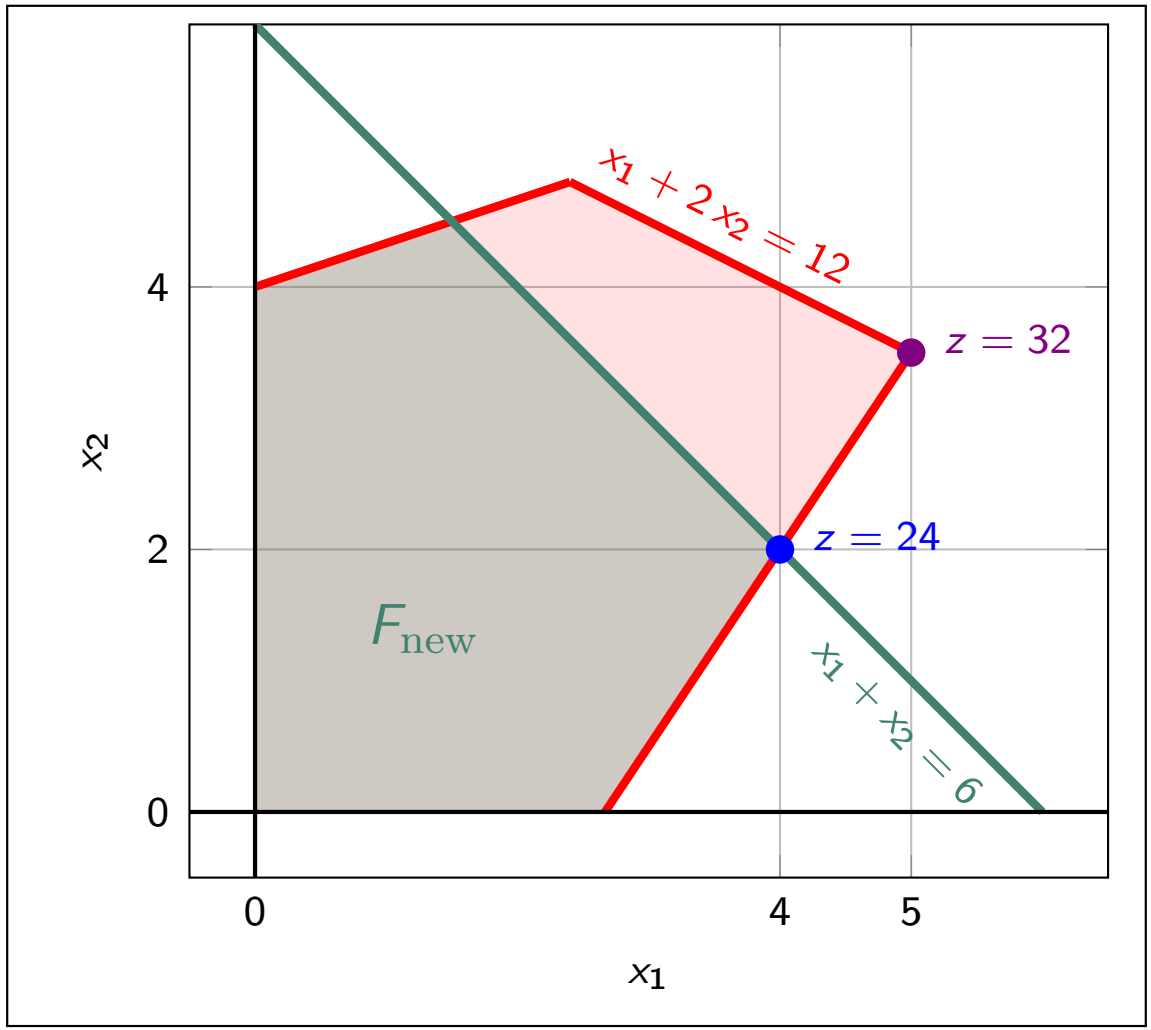
- The ratio test gives

$$\min \left\{ \underbrace{\frac{-2}{-5/8}}_{=\frac{16}{5}}, \underbrace{\frac{-1}{-1/8}}_{=8} \right\} = \frac{16}{5},$$

  and then the pivot column is x_4
- After this single dual pivoting operation the basis becomes primal feasible
- The new optimal solution is

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \qquad \text{with} \qquad z(\,LP\,) = 5 \cdot 4 + 2 \cdot 2 = 24.$$

Example: adding a new constraint (5/6)



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Example: adding a new constraint (6/6)

- In the figure, the new constraint $x_1 + x_2 = 6$ cuts the old feasible region: the new feasible region F_{new} is drawn in green
- In violet the optimal solution of the original LP problem, which violates the new constraint
- In blue the optimal solution of the new LP problem, reached with one dual pivoting operation

Summary

New constraint $\sum_j a_{\star j} x_j \leq b_\star$ with slack variable x_{n+m+1}:

- The extended basis is still dual feasible
- Its primal feasibility depends on $\tilde{\boldsymbol{x}}(\, LP\,)$

Two cases:

- $\tilde{\boldsymbol{x}}(\, LP\,)$ satisfies the new constraint: the optimal solution does not change
- $\tilde{\boldsymbol{x}}(\, LP\,)$ violates it: the dual simplex algorithm restarts from the extended basis, with x_{n+m+1} as the first pivot row

Example: $\bar{b}_6 = -\frac{5}{2}$, one dual pivoting operation, $z(\, LP\,) = 24$

Summary of the chapter (1/2)

Given an optimal basis $\boldsymbol{A}_{\mathcal{B}}$ of an LP problem, a change of its input data is handled as follows:

change	**condition to check**	**how to restart**
c_k with $k \in \mathcal{N}$	dual feasibility	simplex algorithm
c_k with $k \in \mathcal{B}$	dual feasibility	simplex algorithm
b_k	primal feasibility	dual simplex algorithm
new variable $x_\star$	dual feasibility	simplex algorithm
new constraint	primal feasibility	dual simplex algorithm

Summary of the chapter (2/2)

Warm start:

- In all the cases the previous optimal basis is used as a **warm start**
- The two algorithms restart from it, and usually a small number of pivoting operations is enough to restore optimality

Intervals and shadow prices:

- In the interval of a basic coefficient c_k the optimal solution does not change and $z(\,LP\,)$ becomes $z(\,LP\,) + \delta\,\tilde{x}_k$
- In the interval of b_k the optimal value becomes $z(\,LP\,) + \delta\,\pi_k$: the shadow price π_k is the marginal value of the k-th constraint